Tutorial 5: Local Density and Clustering

Three required tasks

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How to use this tutorial

- Attempt each problem before reading its solution.
- Write the main calculation or argument clearly.
- Explain what the result means, not only the final number or formula.
- Complete Tasks 1–3 first. The final task is an optional challenge.

Task 1: KDE and bandwidth selection

Consider the observations:

$$x_1 = 0, \qquad x_2 = 2, \qquad x_3 = 3.$$

Use the triangular kernel:

$$K(u) = \begin{cases} 1 - |u|, & |u| \le 1, \\ 0, & |u| > 1. \end{cases}$$

The KDE is:

$$\hat{p}_h(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right).$$

1. Compute $\hat{p}_h(2)$ for $h = 1$ and $h = 2$.
2. Compute $\hat{p}_h(1)$ for $h = 1$ and $h = 2$.
3. A validation observation is $x^{\text{val}} = 1$. Which bandwidth assigns it the larger density?
4. Based on these calculations, describe what changing h changes in the KDE.

Task 2: Mean-shift iterations

Consider:

$$x_1 = 0, \qquad x_2 = 1, \qquad x_3 = 4, \qquad x_4 = 5.$$

Use Gaussian mean-shift weights with $h = 1$:

$$w_i(x) = \exp\left(-\frac{(x - x_i)^2}{2}\right).$$

Start from:

$$x^{(0)} = 2.$$

1. Compute the four weights $w_i(x^{(0)})$.
2. Compute the local weighted mean $m(x^{(0)})$.
3. Compute the mean-shift vector $s(x^{(0)})$ and obtain $x^{(1)}$.
4. Recompute the weights at $x^{(1)}$ and obtain $x^{(2)}$.
5. From the two updates, identify which group the trajectory is moving toward.

You may use a calculator and report numerical values to three decimal places.

Task 3: Graph cuts and the Laplacian

Consider:

$$\boldsymbol{W} = \begin{bmatrix} 0 & 3 & 0 & 0 \\ 3 & 0 & 1 & 0 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 3 & 0 \end{bmatrix}.$$

Compare:

$$\mathcal{P}_1 : \{1, 2\} \mid \{3, 4\}, \qquad \mathcal{P}_2 : \{1\} \mid \{2, 3, 4\}.$$

1. Compute the degree matrix $\boldsymbol{D}$ and Laplacian $\boldsymbol{L} = \boldsymbol{D} - \boldsymbol{W}$.
2. Compute $\mathrm{cut}(A, B)$ for both partitions.
3. Compute $\mathrm{Ncut}(A, B)$ for both partitions.
4. For $\boldsymbol{f} = [1, 1, 0, 0]^\top$, compute $\boldsymbol{f}^\top \boldsymbol{L} \boldsymbol{f}$.
5. Determine which partition is preferred by normalized cut.

Tutorial 5 complete

Questions and discussion